

Question	Marking Guidance	Mark	Comments
10.1	1. Polymer of nucleotides; 2. Each nucleotide formed from deoxyribose, a phosphate (group) and an organic/nitrogenous base; 3. Phosphodiester bonds (between nucleotides); 4. Double helix/2 strands held by hydrogen bonds; 5. (Hydrogen bonds/pairing) between adenine, thymine and cytosine, guanine;	5	1. Accept 'Polynucleotide' 1. Accept for 'phosphate'. phosphoric acid

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10.2	1. (Simple) diffusion of small/non-polar molecules down a concentration gradient; 2. Facilitated diffusion down a concentration gradient via protein carrier/channel; 3. Osmosis of water down a water potential gradient; 4. Active transport against a concentration gradient via protein carrier using ATP; 5. Co-transport of 2 different substances using a carrier protein;	5	For any answer accept a correct example If no reference to 'small/non-polar' for 1. accept this idea from 'large/charged' given in description of 2. 2. Reject if active rather than passive 5. For 'carrier protein' accept symport OR co-transport protein

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10.3	1. Magnification (figures) show A is bigger than B; 2. A has a nucleus whereas B has free DNA; 3. A has mitochondria whereas B does not; 4. A has Golgi body/endoplasmic reticulum whereas B does not; 5. A has no cell wall whereas B has a murein/glycoprotein cell wall; 6. A has no capsule whereas B has a capsule; 7. A has DNA is bound to histones/proteins whereas B has DNA not associated with histones/proteins OR A has linear DNA whereas B has circular DNA; 8. A has larger ribosomes;	5 max	Accept in all marking points, animal/eukaryote for A and prokaryote/bacterium for B 5. Accept peptidoglycan